Information Extraction: Relation Extraction (cont)

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Text Processing
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Learning objectives

- Define and Explain Bootstrapping and Distant Supervision approaches for relation extraction
- Explain the DIPRE approach for relation extraction
- Name open challenges in RE research

[Information Extraction] Relation Extraction

Information Extraction: Overview

- Introduction to Information Extraction
  - Definition + contrast with IR
  - Example Applications
  - Overview of Tasks
  - Overview of Approaches
  - Evaluation + Shared Task Challenges
  - Brief(est) history of IE
- Named Entity Recognition
  - Task
  - Approaches: Rule-based, Supervised Learning
  - Entity Linking
- **Relation Extraction**
  - **Task**
  - **Approaches: Rule-based, Supervised Learning, Bootstrapping, Distant Supervision**

Information Extraction: Relation Extraction – recap

Relation Extraction (RE)

Task: Identify all **assertions of relations** holding between **entities** in a text T

- the entities are identified in previous entity extraction step
- the set of possible relations $\mathbf{R}$ is determined in advance

Information Extraction: Relation Extraction – recap

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Approaches:

- Knowledge engineering
  - → **shallow**: pattern-action rules
  - → **deep**: parse tree transduction rules

Information Extraction: Relation Extraction – recap

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Approaches:

- Knowledge engineering
  - → **shallow**: pattern-action rules
  - → **deep**: parse tree transduction rules
- Supervised learning
  - → classifies whether a **specific relation** is hold between entities
  - → features from NEs, words between NEs, etc.

Information Extraction: Relation Extraction: Overview

Relation Extraction

- Task definition

Approaches to Relation Extraction

- Knowledge-engineering approaches to RE
- Supervised learning approaches to RE
- **Bootstrapping Approaches to RE**
- Distant Supervision Approaches to RE

Information Extraction: Relation Extraction: Bootstrapping

Motivation: reduce the number of manually labelled examples needed to build a system

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Requirements:

1. a document collection $\mathcal{D}$
2. set of trusted tuples **T**, also called **seed tuples**
3. set of trusted patterns **P**, also called **seed patterns**

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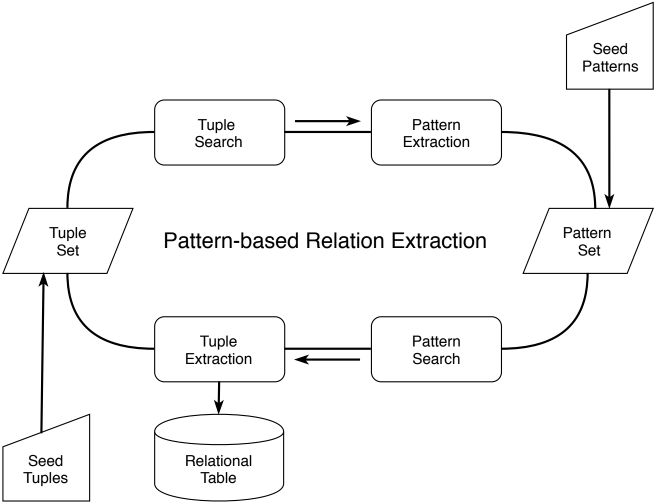
Principle:

1. Find tuples from **T** in $\mathcal{D} \Rightarrow$ extract patterns, add them to **P**
2. Match patterns **P** in $\mathcal{D} \Rightarrow$ extract tuples, add them to **T**

$\rightarrow$ Rinse, repeat

Information Extraction: Relation Extraction: Bootstrapping



[Source: Jurafsky and Martin, 2nd ed., p. 740]

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

DIPRE Dual Iterative Pattern Relation Expansion – proposed by Sergey Brin (1999) [Brin, 1999]

Aim: to extract useful relational tuples from the Web, of the form ⟨PERSON, BOOK_TITLE⟩ Ex.: ⟨LEO TOLSTOY, WAR AND PEACE⟩

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Method:

- Exploit **duality of patterns and relations**
  - Good tuples help find good patterns
  - Good patterns help find good tuples
- Use the bootstrapping method starting with **user-supplied tuples**

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

Isaac Asimov	The Robots of Dawn
Charles Dickens	Great Expectations
Leo Tolstoy	War and Peace
Jane Austen	Pride and Prejudice
Marcel Proust	In Search of Lost Time

Main loop in DIPRE is the following:

1. $\mathcal{R}' \leftarrow$ Seed tuples

   $\mathcal{R}'$ contains sample instances of the target relation

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   $\mathcal{P}$ contains the patterns generated based on the occurrences
   Seek for low error rate patterns, ideally having high coverage

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5. Stop if $\mathcal{R}'$ is large enough, otherwise go to 2

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

2. $\mathcal{O} \leftarrow \text{FindOccurrences}(\mathcal{R}', \mathcal{D})$

   $\mathcal{O}$ contains all occurrences of tuples in $\mathcal{R}'$ appearing in $\mathcal{D}$

- **Occurrences** are defined as 7-tuples **⟨prefix, author, middle, title, suffix, order, url⟩**
  - **order** is **true** for ⟨author, title⟩ tuples, **false** for ⟨title, author⟩ tuples
  - **url** is the document url containing the occurence
  - **prefix** is the *m* characters (in tests m=10) preceding the author (or title)
  - **middle** is the text between author and title
  - **suffix** is the *m* characters following the title (or author)

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

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- Ex. search for **⟨Charles Dickens, Great Expectations⟩** in the domain **www.books.com**
  - → URL: **www.books.com/TopRated**
  - → text: "The famous writer Charles Dickens wrote Great Expectations book"

[Source: https://cs.uwaterloo.ca/~kmsalem/courses/CS848W10/presentations/Anup-proj.pdf]

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  - $\rightarrow$ URL: **www.books.com/TopRated**
  - $\rightarrow$ text: "The famous writer Charles Dickens wrote Great Expectations book"
- Extracted occurrence:

  ⟨The famous writer, Charles Dickens, wrote, Great Expectations, book, true, www.books.com/TopRated⟩

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Information Extraction: Relation Extraction: Bootstrapping - DIPRE

3. $\mathcal{P} \leftarrow$ GenPatterns($\mathcal{O}$)

- **Patterns** are defined as 5-tuples **⟨prefix, middle, suffix, order, urlprefix⟩**
  - **order** is **true** for ⟨author, title⟩ tuples, **false** for ⟨title, author⟩ tuples
  - the tuple matches the pattern if
    - there is a document in the collection (web) with URL matches **urlprefix**,
    - that contains text matching the RegEx /.*prefix author middle title suffix.*/

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

3. $\mathcal{P} \leftarrow$ GenPatterns($\mathcal{O}$)

- Group occurrences having similar **order** and **middle**

  ⟨The famous writer, Charles Dickens, **wrote**, Great Expectations, book, **true**, www.books.com/TopRated⟩

  ⟨The great writer, Nicholas Sparks, **wrote**, The Last Song, book, **true**, www.books.com/BestSellers⟩

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- Generate a pattern as general as possible:

  /**writer .*? wrote .*? book**/ with **order**=true and **urlprefix**=www.books.com

[Source: https://cs.uwaterloo.ca/~kmsalem/courses/CS848W10/presentations/Anup-proj.pdf]

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

4. $\mathcal{R}' \leftarrow M_{\mathcal{D}}(\mathcal{P})$

   Update $\mathcal{R}'$ with the tuples from $\mathcal{D}$ matching patterns in $\mathcal{P}$

/writer .*? wrote .*? book/ with **order**=true and **urlprefix**=www.books.com

- Result: ......The writer **Mario Puzo** wrote **The Godfather** book.....

⇒ Extract relation ⟨**Mario Puzo, The Godfather**⟩

[Source: https://cs.uwaterloo.ca/~kmsalem/courses/CS848W10/presentations/Anup-proj.pdf]

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

4. Stop if $\mathcal{R}'$ is large enough, otherwise go to ❷ $\mathcal{O} \leftarrow$ FindOccurrences($\mathcal{R}'$, $\mathcal{D}$)

- New tuple: **⟨Mario Puzo, The Godfather⟩**
- New match: .... the book **The Godfather** was written by **Mario Puzo**....
- New occurrence: ⟨the book, Mario Puzo, was written by, The Godfather, NULL, **false**, www.library.com⟩
- New pattern: ⟨The book, was written by, NULL, false, www.library.com⟩

/the book .*? was written .*?/ with **order**=false and **urlprefix**=www.library.com

[Source: https://cs.uwaterloo.ca/~kmsalem/courses/CS848W10/presentations/Anup-proj.pdf]

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

- Experiment on 24 million web pages

→ **only** 5 seed tuples !

Isaac Asimov	The Robots of Dawn
David Brin	Startside Rising
James Gleick	Chaos: Making a New Science
Charles Dickens	Great Expectations
William Shakespeare	The Comedy of Errors

Information Extraction: Relation Extraction: Bootstrapping - DIPRE

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1. 1st iteration $\rightarrow$ 199 occurrences and 3 patterns
   $\rightarrow$ 4,047 unique $\langle$author, title$\rangle$ tuples

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2. 2nd iteration over 5 millions web pages $\rightarrow$ 3,972 occurrences
   $\rightarrow$ 105 patterns
   $\rightarrow$ 9,369 $\langle$author, title$\rangle$ tuples
   - only manual action: some bad authors ("Conclusion") were rejected

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   - only manual action: some bad authors ("Conclusion") were rejected
3. Final iteration → 9,988 occurrences → 346 patterns
   → 15,257 (almost) unique ⟨author, title⟩ tuples

⇒ list available here:
`http://infolab.stanford.edu/~sergey/booklist.html`

Information Extraction: Relation Extraction: Bootstrapping

- **Strengths**
  - No need for manually annotated data

  $\rightarrow$ only a few examples are required

Information Extraction: Relation Extraction: Bootstrapping

- **Strengths**
  - No need for manually annotated data
  - $\rightarrow$ only a few examples are required
- **Weaknesses**
  - Can suffer from **semantic drift**
    - $\rightarrow$ an erroneous pattern introduces erroneous tuples leading to erroneous patterns...
    - a confidence score on patterns and tuples can mitigate this problem
  - Works well only when significant redundancy in expressing a relation
  - Issues when multiple relations holds between the same pair of entities
    - tuple $\langle$name, place of birth$\rangle$ could be mixed up with $\langle$name, place of death$\rangle$
    - a sentence containing an occurrence of one of those tuple could express any of the two relations

Information Extraction: Relation Extraction: Overview

Relation Extraction

- Task definition

Approaches to Relation Extraction

- Knowledge-engineering approaches to RE
- Supervised learning approaches to RE
- Bootstrapping Approaches to RE
- **Distant Supervision Approaches to RE**

Information Extraction: Relation Extraction: Distant Supervision

Distant supervision also referred to as **lightly supervised** machine learning

Aim: reduce/eliminate the need for manually labelled data
Principle: use a structured data source $\mathcal{R}$ (e.g. a database) to label a large document collection $\mathcal{D}$ then train a standard supervised ML system

Information Extraction: Relation Extraction: Distant Supervision

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Principle: use a structured data source $\mathcal{R}$ (e.g. a database) to label a large document collection $\mathcal{D}$ then train a standard supervised ML system

1. Search for sentences in $\mathcal{D}$ containing the entity pairs that occurs in relation instances (tuples) in $\mathcal{R}$
2. Label these sentences as positive occurrences of the relation
3. Use the labelled sentences as training data for a standard supervised relation extractor (ML algorithm)

Information Extraction: Relation Extraction: Distant Supervision

- Well known approach: Mintz et al. (2009)
- They use Freebase as structured data source

Relation name	Size	Example
/people/person/nationality	281,107	John Dugard, South Africa
/location/location/contains	253,223	Belgium, Nijlen
/people/person/profession	208,888	Dusa McDuff, Mathematician
/people/person/place of birth	105,799	Edwin Hubble, Marshfield
/dining/restaurant/cuisine	86,213	MacAyo's Mexican Kitchen, Mexican
/business/business chain/location	66,529	Apple Inc., Apple Inc., South Park, NC
/biology/organism classification rank	42,806	Scorpaeniformes, Order
/film/film/genre	40,658	Where the Sidewalk Ends, Film noir
/film/film/language	31,103	Enter the Phoenix, Cantonese
/biology/organism higher classification	30,052	Calopteryx, Calopterygidae
/film/film/country	27,217	Turtle Diary, United States
/film/writer/film	23,856	Irving Shulman, Rebel Without a Cause
/film/director/film	23,539	Michael Mann, Collateral
/film/producer/film	22,079	Diane Eskenazi, Aladdin
/people/deceased person/place of death	18,814	John W. Kern, Asheville
/music/artist/origin	18,619	The Octopus Project, Austin
/people/person/religion	17,582	Joseph Chartrand, Catholicism
/book/author/works written	17,278	Paul Auster, Travels in the Scriptorium
/soccer/football position/players	17,244	Midfielder, Chen Tao
/people/deceased person/cause of death	16,709	Richard Daintree, Tuberculosis
/book/book/genre	16,431	Pony Soldiers, Science fiction
/film/film/music	14,070	Stavisky, Stephen Sondheim
/business/company/industry	13,805	ATS Medical, Health care

[Source: Mintz et al. (2009) [Mintz et al., 2009]]

Information Extraction: Relation Extraction: Distant Supervision

- **Freebase**: free online database of structured semantic data
  - data from Wikipedia infoboxes + other open access sources
  - 102 relations connecting 940,000 entities $\rightarrow$ 1.8 millions instances
  - Freebase no longer available: bought by Google $\rightarrow$ Google Knowledge Graph (partly free / partly paid access)
  - Similar sources: **DBPedia** and **WikiData**

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- Mintz et al. (2009) use a dump of Wikipedia as their **document collection**
  - $\sim$ 1.8 million articles with 14.3 sentences/article in average
  - 800,000 articles for training / 400,000 for testing

Information Extraction: Relation Extraction: Distant Supervision

Distant supervision

Assumption: any sentence containing two entities that participate in a relation might express that relation → tag those sentences as mentions of the relation

Information Extraction: Relation Extraction: Distant Supervision

Distant supervision

Assumption: any sentence containing two entities that participate in a relation might express that relation $\rightarrow$ tag those sentences as mentions of the relation

- Same relation may be expressed in different ways in different sentences:

  [Steven Spielberg]$_{PER}$'s film [Saving Private Ryan]$_{FILM}$ is loosely based on the brothers' story.

  Allison co-produced the Academy Award-wining [Saving Private Ryan]$_{FILM}$, directed by [Steven Spielberg]$_{PER}$.

Information Extraction: Relation Extraction: Distant Supervision

Distant supervision

Assumption: any sentence containing two entities that participate in a relation might express that relation $\rightarrow$ tag those sentences as mentions of the relation

- Combine features from multiple mentions to get richer feature vector

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- Combine features from multiple mentions to get richer feature vector
- Use multiclass logistic regression as a machine learning framework

Information Extraction: Relation Extraction: Distant Supervision

Distant supervision

Assumption: any sentence containing two entities that participate in a relation might express that relation $\rightarrow$ tag those sentences as mentions of the relation

- Combine features from multiple mentions to get richer feature vector
- Use multiclass logistic regression as a machine learning framework
- At test time: combine all occurrences of a given entity pair, assign the most likely relation (or none)

Information Extraction: Relation Extraction: Distant Supervision

- Need for **negative instance** $\rightarrow$ an 'unrelated' relation
  - Randomly select entity pairs not appearing in Freebase relations, extract features from them
  - Rare cases: the relation could be wrongly omitted from Freebase $\rightarrow$ low effect on performance

Information Extraction: Relation Extraction: Distant Supervision

- Need for **negative instance** $\rightarrow$ an 'unrelated' relation
  - Randomly select entity pairs not appearing in Freebase relations, extract features from them
  - Rare cases: the relation could be wrongly omitted from Freebase $\rightarrow$ low effect on performance
- Evaluation
  - Human evaluation of highest 100 and 1000 results per relation for 10 relations
  - Avg. precision for best feature combinations: 69% for top 100 and 68% for top 1000
  - Competitive results with knowledge engineering and supervised learning methods

Information Extraction: Relation Extraction: Distant Supervision

- **Strengths**
  - Need for manually labelled training data is eliminated

  $\rightarrow$ still need some expert linguistic resources (e.g. Freebase)
  - Can very rapidly get extractors for a wide range of relations

Information Extraction: Relation Extraction: Distant Supervision

- **Strengths**
  - Need for manually labelled training data is eliminated
  - $\rightarrow$ still need some expert linguistic resources (e.g. Freebase)
  - Can very rapidly get extractors for a wide range of relations

- **Weaknesses**
  - Precision still a bit behind best knowledge-engineering/supervised ML approaches
  - Requires an as large as possible structured database for the relations of interest.

Information Extraction: Relation Extraction: Conclusion

Relation extraction

aims to **detect** and **classify** all mentions of a **given set of relations** holding between **specified entities** within a given text

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Information Extraction: Relation Extraction: Conclusion

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- Presented four different approaches:
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Information Extraction: Relation Extraction: Conclusion

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- Core IE technology, **very difficult** due to the **high variabilities** of relation expressions in natural language
- Presented four different approaches:
  - Knowledge engineering, supervised machine learning, bootstrapping and distant supervision
- **Open challenges**
  - improve precision and recall
  - hande relations expressed over several sentences
  - handle textual entailment
  - improve bootstrapping methods to avoid the **semantic drift**
  - develop relation extractors for languages other than English (e.g. under-resourced languages)

Reading material

- Daniel Jurafsky & James H. Martin (2021).
  Chapter 17: Information Extraction.
  Speech and Language Processing, 3rd ed. draft.
  `https://web.stanford.edu/~jurafsky/slp3/17.pdf`
- Brin, S. (1999).
  Extracting patterns and relations from the world wide web.
  In Selected Papers from the International Workshop on The World Wide Web and Databases, WebDB '98, pages 172–183, London, UK. Springer-Verlag
  `http://ilpubs.stanford.edu:8090/421/1/1999-65.pdf`

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

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